

## **MEMORANDUM**

**To:** Bay Area Rapid Transit District Board of Directors

**From:** Jerra McLaughlin and Mina Mohebbi

**Topic:** Investments in Public Transportation

**Date:** December 5, 2024

### **Executive Summary**

This analysis examines whether the BART expansion to Warm Springs in Fremont, CA, led to an increase in local property values in the two closest zip codes (94538 and 94539).<sup>1</sup> Using a Weighted Least Squares (WLS) regression model with a one-month lag and controls for mortgage rates and unemployment rates, we found no statistically significant effect of the transit center's opening on property values, as the interaction term's p-value was greater than the 0.05 significance level. However, mortgage rates and unemployment rates exhibited significant relationships with property values, indicating broader economic factors played a key role. Due to limitations of zip-code level data and potential omitted variable bias, the precision and generalizability of the findings are restricted. To better evaluate the impacts of transit investments, policymakers should prioritize access to more granular data, support transit-oriented development (TOD) policies, and consider broader economic and housing affordability factors in future transit projects.

### **Research Question and Background**

Property values have generally been shown to increase as a result of transit projects, according to the Federal Transit Administration.<sup>2</sup> Increasing the connections, reliability, and convenience of public transportation within a region can support value creation.<sup>3</sup> We sought to examine the impact of the BART expansion in the East Bay on property values within a region already recognized as one of the most expensive housing markets in California and the United States, both for home purchases and rentals.<sup>4</sup>

The 2008 recession and California budget crisis delayed the Warm Springs project, pushing construction from its original start date to 2009 and extending the timeline. Initially slated to open in 2014, the station began service on March 25, 2017.<sup>5</sup> Contributing factors included reduced transportation funding during the state budget crisis and the dissolution of redevelopment agencies in 2011.<sup>6</sup> To address budget shortfalls, the Metropolitan Transportation Commission redirected \$91 million from the Dumbarton Rail Corridor project to the Warm Springs project in 2008.<sup>7</sup> Additionally, the 2016 passage of California's SB-167 (Housing Accountability Act) aimed to facilitate housing development near transit but likely impacted housing approvals only after the station's opening.<sup>8</sup> To account for these delays and broader market corrections post-recession, the analysis uses a timeline from 2011 (three years before the originally anticipated opening) to 2020 (three years after the actual opening).

## **Empirical Model**

Our empirical model assesses the impact of the Warm Springs BART station on local property values. The dependent variable is the logarithmically-transformed, inflation-adjusted property value. Key independent variables include a treatment indicator for the two nearest zip codes (94538 and 94539), a post-treatment indicator for the period after the station's opening, and their interaction term, which captures the relative effect of the transit center on the treatment group. Control variables include mortgage rates, unemployment rates, and lagged property values to account for broader economic trends and market conditions. The model helps estimate whether the transit investment led to a measurable change in property values while controlling for other influencing factors.

## **Data, Limitations, and Methodology**

Our data was retrieved from the publicly available Zillow Research API, American Community Survey 5-year estimates from the U.S. Census, the U.S. Bureau of Labor Statistics, and SmartAsset.<sup>9 10 11 12</sup>

The "Zillow Home Value Index" (ZHVI) is utilized in our model as a measure of property value. This value is described as "a measure of the typical home value and market changes across a given region and housing type. It reflects the typical value for homes in the 35th to 65th percentile range. Available as a smoothed, seasonally adjusted measure and as a raw measure."<sup>13</sup>

We enhanced our dataset by incorporating additional variables such as median household income, household size, unemployment rates, mortgage rates, and housing units. These variables provide greater depth to the analysis by allowing us to examine the economic and demographic factors influencing property values in the study areas. These measures of median household income and housing units are from ACS 5-year Estimate Subject Tables, specifically S1903 (Median Income in past 12 months), and DP05 (Demographic and Housing Estimates), respectively.<sup>14 15</sup> Our measure of unemployment rates came from the U.S Bureau of Labor Statistics, and mortgage rates from SmartAsset. We used CPI from FRED to adjust the ZHVI values to 2020 inflation-adjusted dollars.<sup>16</sup>

We employed a Weighted Least Squares (WLS) regression model, incorporating a one-period lag (equivalent to one month in this analysis) as an explanatory variable. We regressed log-transformed property values on the two binary variables, their interaction, the lagged log-transformed property value and two controls: mortgage rates and unemployment rates. Unemployment rates are a proxy for the economic conditions in the areas, and demand for housing as high unemployment would likely result in a low demand for housing.

## **Findings and Interpretation**

Based on our findings, the construction and completion of the Warm Springs Transit Center has no effect on local property values in the two zip codes in closest proximity to the Warm Springs Transit Center. The coefficient of interest, the interaction between `treatment_group` and `post_treatment`, is -0.0584, but the p-value is greater than the

significance level ( $\alpha$ ) = 0.05, at 0.201. This indicates that the effect is not statistically significant, and we cannot definitively conclude that the investment in public transportation at Warm Springs had a statistically significant effect on property values in the treatment group during the post-treatment period.

The variables `mortgage_rate` and `unemployment_rate` did exhibit statistically significant relationships with property value, since  $p < \alpha = 0.05$ . Specifically, since we log-transformed the data, we can interpret the coefficients as percentage changes: an approximate 1% increase in mortgage rates leads to an approximate 10% decrease in property value, and a 1% increase in unemployment rates corresponds to an approximate 4% decrease in property value in the zip codes in this dataset.

However, we cannot confidently generalize these results beyond the zip codes analyzed. The model may suffer from omitted variable bias, as we could not include variables such as median household income, household size, or the number of households due to multicollinearity issues. This limitation suggests that the estimated coefficients may be artificially larger than what might be observed in the real world.

## **Policy Recommendations and Conclusion**

To improve the precision of future analyses and better understand the impacts of transit expansions, policymakers should prioritize broader access to granular data. Access to property-level or block-level data, rather than aggregated zip code-level data, would allow for a more accurate examination of localized effects. Policymakers could help with this by encouraging collaboration between local governments, county assessor's offices, and organizations like Zillow or real estate developers to share detailed data on property transactions, zoning changes, and economic activity near transit stations.

Although our findings did not show a statistically significant relationship between the Warm Springs Transit Center and property values, existing research suggests that transit investments often create long-term value. Policymakers should take advantage of these opportunities by integrating transit-oriented development (TOD) policies into future projects.<sup>17</sup> High-density, mixed-use developments near transit stations should be incentivized, particularly in underutilized or commercially zoned areas, to maximize economic and housing benefits.<sup>18</sup> To address potential affordability challenges due to property values near transit, local governments should adopt measures such as inclusionary zoning to ensure affordable housing remains available.<sup>19</sup> This is something we had hoped we could propose with more concrete findings, especially with respect to the specific locality we were analyzing.

Beyond property values, the broader economic impacts of transit investments should be evaluated. Policymakers could look at how transit expansions affect job accessibility, emissions, and commercial activity to ensure these projects are meeting their intended goals. Future research into these areas could guide transit-related policy decisions and justify further investments and extensions of existing transit networks.

## Endnotes

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